WAKISSHA UACE CHEMISTRY 2 PROPOSED SCORING GUIDE 2026

By:Dr.Tushabe Derrick

ITEM 1: DETERMINATION OF THE PURITY OF SODIUM NITRITE CRYSTALS

Aim

To determine the percentage purity of the supplied sodium nitrite crystals by back titration and establish whether they meet the required minimum purity of 98%.

Hypothesis

The supplied sodium nitrite crystals are expected to have a purity of about 98% or above, making them suitable for the intended purpose.

Variables

Independent variable

Volume of ammonium ferrous sulphate solution used to titrate the excess potassium manganate(VII).

Dependent variable

Percentage purity of the sodium nitrite crystals.

Controlled variables

- Volume of potassium manganate(VII) used.
- Volume of sodium nitrite solution used.
- Concentration of ammonium ferrous sulphate.
- Volume and concentration of sulphuric acid.
- Temperature.
- Time allowed for the reaction between nitrite ions and manganate(VII) to go to completion.

Apparatus

Burette, pipette, pipette filler, conical flask, 250 cm^3 volumetric flask, beaker, measuring cylinder, funnel, electronic balance, white tile, wash bottle and glass rod.

Reagents

- Sodium nitrite crystals, Q
- Potassium manganate(VII), FA1
- 0.20 M ammonium ferrous sulphate, FA2
- 2 M sulphuric acid
- Distilled water

Procedures

Preparation of sodium nitrite solution

1. 0.80 g of sodium nitrite crystals Q was accurately weighed.
2. The crystals were dissolved in distilled water in a beaker.
3. The solution was quantitatively transferred into a 250 cm^3 volumetric flask.
4. The beaker was rinsed several times with distilled water and the washings were added to the volumetric flask.
5. Distilled water was added up to the calibration mark.
6. The flask was stoppered and shaken thoroughly.

Standardisation of potassium manganate(VII)

7. The burette was rinsed with FA1 and filled with potassium manganate(VII).
8. 25.0 cm^3 of FA1 was pipetted into a clean conical flask.

9. An equal volume of 2 M sulphuric acid was added.
10. The acidified FA1 was titrated with 0.20 M ammonium ferrous sulphate.
11. The titration was continued until the purple colour of potassium manganate(VII) just disappeared.
12. The titration was repeated until concordant titres were obtained.

Back titration

13. The pipette was rinsed with the prepared sodium nitrite solution.
14. 25.0 cm^3 of the sodium nitrite solution was pipetted into a clean conical flask.
15. 25.0 cm^3 of standardised potassium manganate(VII) was added.
16. An equal volume of 2 M sulphuric acid was added.
17. The mixture was allowed sufficient time for the reaction to go to completion.
18. The excess potassium manganate(VII) was titrated with 0.20 M ammonium ferrous sulphate.
19. The end point was reached when the purple colour of potassium manganate(VII) just disappeared.
20. The titration was repeated to obtain concordant titres.

Data presentation

Table 1: Standardisation of potassium manganate(VII)

Titration	Initial burette reading (cm^3)	Final burette reading (cm^3)	Titre (cm^3)
Rough	0.00	19.80	19.80
I	0.00	19.50	19.50
II	0.00	19.55	19.55
III	0.00	19.60	19.60

Values used to calculate average volume

$19.50, 19.55, 19.60$

Mean titre:

$(19.50 + 19.55 + 19.60) \div 3$

$= 19.55\ cm^3$

NB: use any two and divide by 2

Table 2: Back titration

Titration	Initial burette reading (cm^3)	Final burette reading (cm^3)	Titre (cm^3)
Rough	0.00	8.60	8.60
I	0.00	8.15	8.15
II	0.00	8.20	8.20
III	0.00	8.20	8.20

Values used to calculate average volume

$8.15, 8.20, 8.20$

Mean titre:

$(8.15 + 8.20 + 8.20) \div 3$

$= 8.18\ cm^3$

NB: use any two and divide by , preferably 8.20 and 8.20

Data analysis

The reactions involved are:

$2MnO_4^- + 5NO_2^- + 6H^+ \rightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$

$MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

A. Standardisation of potassium manganate(VII)

1. Amount of ammonium ferrous sulphate in 19.55 cm³

1000 cm³ of 0.20 M ammonium ferrous sulphate contains:

0.20 mol

Therefore, 19.55 cm³ contains:

$(19.55 \times 0.20) \div 1000$

$= 0.00391 \text{ mol}$

of Fe^{2+}.

2. Amount of potassium manganate(VII) in 25.0 cm³

From the equation:

$1MnO_4^- : 5Fe^{2+}$

Therefore:

$5 \text{ mol } Fe^{2+} \rightarrow 1 \text{ mol } MnO_4^-$

$0.00391 \text{ mol } Fe^{2+} \rightarrow 0.00391 \div 5 \text{ mol } MnO_4^-$

$= 0.000782 \text{ mol } MnO_4^-$

Thus, 25.0 cm³ of FA1 contains 0.000782 mol MnO_4^-.

3. Concentration of potassium manganate(VII)

25.0 cm³ contains:

0.000782 mol

Therefore, 1000 cm³ contains:

$(0.000782 \times 1000) \div 25.0$

$= 0.03128 \text{ mol}$

Therefore, the concentration of FA1 is:

$0.03128 \text{ mol dm}^{-3}$

B. Back titration

1. Amount of potassium manganate(VII) initially added

25.0 cm³ of FA1 was used.

From the standardisation:

$25.0 \text{ cm}^3 \rightarrow 0.000782 \text{ mol } MnO_4^-$

Therefore:

$\text{Amount of } MnO_4^- \text{ initially added} = 0.000782 \text{ mol}$

2. Amount of ammonium ferrous sulphate used

Average titre = 8.18 cm³

1000 cm³ of 0.20 M ammonium ferrous sulphate contains:

0.20 mol

Therefore, 8.18 cm³ contains:

$(8.18 \times 0.20) \div 1000$

$= 0.001636 \text{ mol } Fe^{2+}$

3. Amount of excess potassium manganate(VII)

From:

$1MnO_4^- : 5Fe^{2+}$

$0.001636 \text{ mol } Fe^{2+} \rightarrow 0.001636 \div 5 \text{ mol } MnO_4^-$

$= 0.0003272 \text{ mol } MnO_4^-$

Thus, excess potassium manganate(VII) = 0.0003272 mol.

4. Amount of potassium manganate(VII) that reacted with nitrite

Initially present:

0.000782 mol

Excess remaining:

0.0003272 mol

Therefore, amount that reacted:

$0.000782 - 0.0003272$

$= 0.0004548 \text{ mol}$

5. Amount of nitrite ions in 25.0 cm³

From:

$2MnO_4^- : 5NO_2^-$

$2 \text{ mol } MnO_4^- \rightarrow 5 \text{ mol } NO_2^-$

$0.0004548 \text{ mol } MnO_4^- \rightarrow (0.0004548 \times 5) \div 2 \text{ mol } NO_2^-$

$= 0.001137 \text{ mol } NO_2^-$

Since $NaNO_2 \rightarrow Na^+ + NO_2^-$, the amount of sodium nitrite in the 25.0 cm³ aliquot is:

$0.001137 \text{ mol } NaNO_2$

6. Amount of sodium nitrite in the whole 250 cm³

25.0 cm³ contains:

0.001137 mol

Therefore, 250 cm³ contains:

$(250 \times 0.001137) \div 25.0$

$= 0.01137 \text{ mol } NaNO_2$

7. Mass of pure sodium nitrite

Relative molecular mass of sodium nitrite:

$NaNO_2 = 23 + 14 + (2 \times 16)$

$= 69$

Therefore:

$1 \text{ mol } NaNO_2 \rightarrow 69 \text{ g}$

$0.01137 \text{ mol } NaNO_2 \rightarrow 0.01137 \times 69 \text{ g}$

$= 0.78453 \text{ g}$

Therefore, the mass of pure sodium nitrite present in the original sample is 0.78453 g.

8. Percentage purity

The original sample had a mass of 0.80 g.

$0.80 \text{ g of sample} \rightarrow 100\%$

$0.78453 \text{ g of pure sodium nitrite} \rightarrow (0.78453 \times 100) \div 0.80$

$= 98.06625\%$

$\text{Percentage purity} = 98.1\%$

Risks and mitigation

Risk	Mitigation
Sulphuric acid is corrosive and can cause burns.	Wear safety goggles, gloves and protective clothing.
Potassium manganate(VII) is a strong oxidising agent and stains skin.	Avoid direct contact and clean spills immediately.
Sodium nitrite is harmful if swallowed.	Do not taste chemicals and wash hands after the experiment.
Broken glassware can cause cuts.	Handle apparatus carefully and report breakages immediately.
Chemical splashes can damage the eyes.	Wear safety goggles throughout the experiment.
Inaccurate burette readings can affect the result.	Read the meniscus at eye level.
The end point may be overshot.	Add the titrant slowly, especially near the end point.

Conclusion

The percentage purity of the supplied sodium nitrite crystals was found to be:

$$98.1\%$$

The required minimum purity is:

$$98.0\%$$

Since:

$$98.1\% > 98.0\%$$

the sodium nitrite crystals meet the required purity standard.

Recommendation

The sodium nitrite crystals are recommended as suitable for the intended purpose, since their experimentally determined purity of 98.1% is above the required minimum of 98%.

However, because the crystals are intended for medical use, the batch should undergo the appropriate official quality-control and regulatory checks before being released for use.